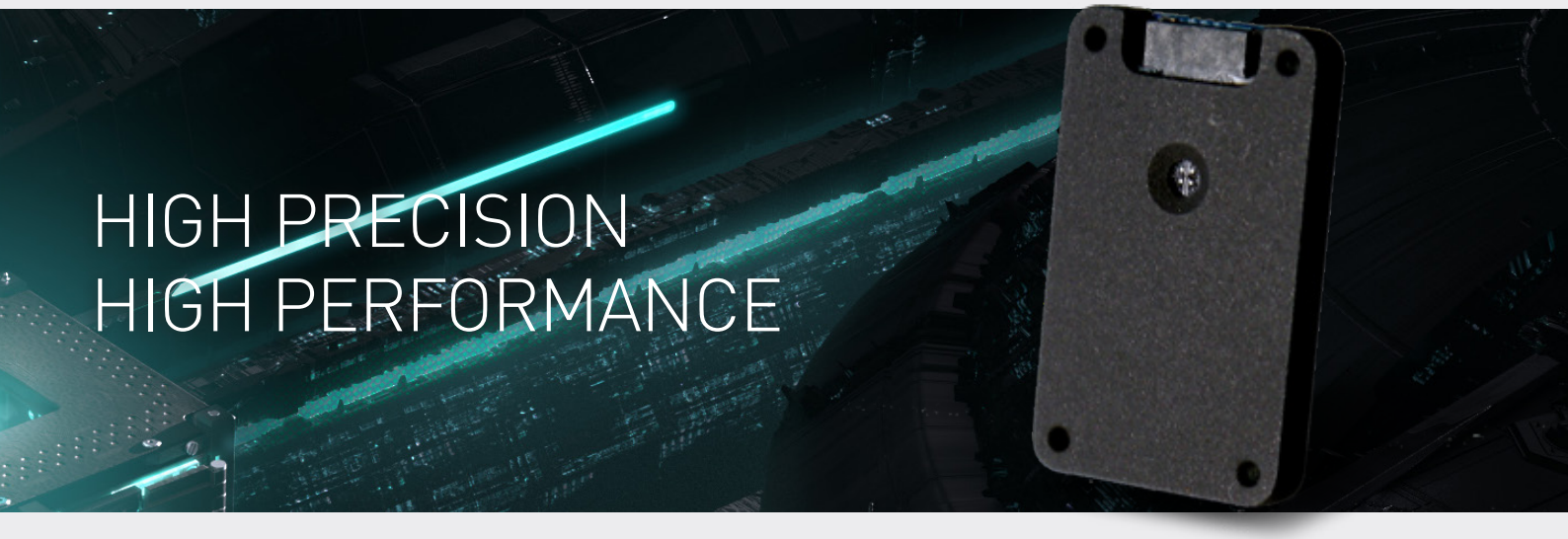
AAC CLYDE SPACE

Sun Sensors

SS200



HIGH PRECISION HIGH PERFORMANCE

The SS200 is a small-size, low-mass and low-power sun sensor for use in satellites. The sampling rate is adjustable, allowing the user to trade bandwidth against power consumption. The SS200 can be seamlessly integrated with AAC Clyde Space's line of integrated attitude determination and control systems to provide a fully integrated ADCS solution. The SS200 delivers 0.3 degree (1-sigma) accuracy in the +/-45 degree range, and is calibrated before delivery.

KEY HIGHLIGHTS:

Mass:	3 g
Power consumption (active)	2.5-40 mW
Sampling rate up to:	100 Hz
Accuracy (+/- 45°) (1-sigma)	<0.3°
Total field of view	110 °
I2C-compatible interface	

COMPATIBILITY

The SS200 can be seamlessly integrated with AAC Clyde Space's line of integrated attitude determination and control systems to provide a fully integrated ADCS solution.

ADAPTABLE

One of SS200's features is that the sampling rate can be adjusted by the customer to suit their needs. The benefits are two-fold: higher sampling rates can be achieved when necessary at a temporary increase in power consumption, while when low sampling rates suffice, power consumption can be optimized.

HERITAGE

The SS200 sun sensors have been flying since 2018 and have been delivered to various clients worldwide.

TECHNICAL SPECIFICATIONS

Performance		
Field of view	110	°
Accuracy (+/- 45° range) (1-sigma)	<0.3	°
Sampling rate	up to 10	Hz

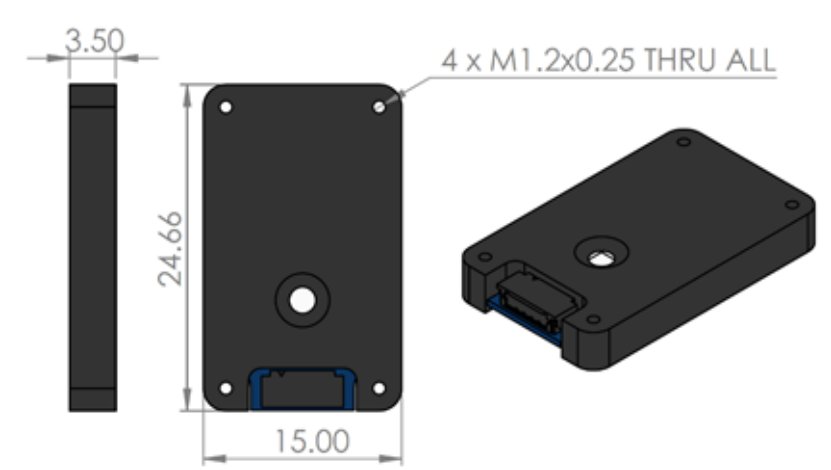
Enviormental		
Operating temperature (electronics)[1]	-45 to +85	°C
Operating temperature (sensor)	-55 to +125	°C
Radiation tolerance	> 36	krad (Si)

Electrical specifications		
Supply voltage	5	V
Power consumption[2] (sampling)	2.5-40	mW
Power required (idle)	< 1.5	mW

Mechanical		
Outer dimensions	24.66 x 15.00 x 3.50	mm
Mass	3	g

1 Temperature range of control electronics. Functionality of mechanical components and sensor extends beyond this range.

2 Power consumption dependent on sampling rate



To make an enquiry, request a quotation or learn about AAC Clyde Space's other products and services, please contact:
enquiries@aac-clydespace.com

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